

Virtual Patient Model

Synthetic EHR and simulated CGM combined to forecast a post-meal glucose spike two hours in advance

36%

demo spike probability

Calculated by the trained model from the canonical synthetic patient

A health record explains the person.

A wearable explains the moment.

A useful virtual patient needs both. Static context changes slowly, while glucose can change within minutes after a meal.

STATIC / HISTORICAL

Age, sex, BMI, diabetes duration, HbA1c, fasting glucose, diagnoses and medication

DYNAMIC / REAL TIME

CGM observations arrive every 15 virtual minutes through the simulated wearable stream

Two-hour glucose spike forecast

03

120

minutes ahead

Adverse event definition

**Glucose rises by at least 40
mg/dL after the current meal**

01

Observe

Read the synthetic patient record and recent CGM history.

02

Forecast

Return a probability, risk band and model factors.

03

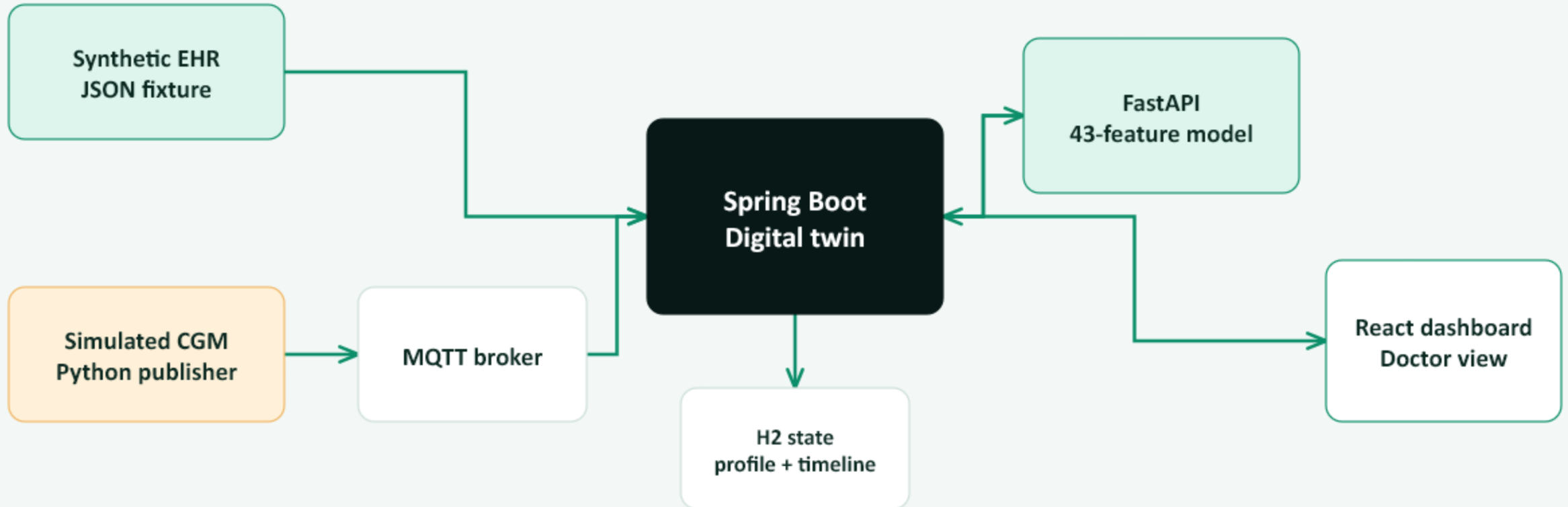
Review

Let the doctor inspect the timeline and refresh the estimate.

A research estimate supports discussion. It does not prescribe treatment.

Static and dynamic data fusion

04



The browser calls Spring Boot only. The model service and MQTT broker stay behind the application boundary.

43

numeric input features

Recent glucose lags and changes carry the short-term signal. Static patient values add clinical context.

CGM history

Current value, 15/30/60/120-minute lags, variation and range

Meal + time

Meal text signals, time of day and minutes since the previous meal

Patient context

Age, sex, BMI, diabetes duration, HbA1c and fasting glucose

Medication context

Recorded insulin and non-insulin medication event flags

Pipeline: median imputation · missingness indicators · standard scaling · Logistic Regression

Patient-level evaluation

06

Each patient belongs to one split only, which reduces leakage between training and evaluation.



0.757

test ROC AUC

These metrics measure prototype discrimination. They do not establish clinical safety, calibration or patient benefit.

Virtual Patient Monitor

SYNTHETIC DEMONSTRATION PATIENT

142

mg/dL current glucose

36%

moderate spike risk

CGM timeline



The working React screen shows the same values through the live Spring Boot API.

- 01 Risk text appears with colour for accessibility.
- 02 Model factors describe influence, not clinical cause.
- 03 Unavailable services never produce a fake score.
- 04 The synthetic and research-only labels stay visible.

PUBLIC DEMO

One fully synthetic patient

The repository includes a generated profile and simulated CGM. It contains no identifiable patient record.

RESEARCH TRAINING DATA

Raw patient-level files stay outside Git

The committed artifact contains only fitted coefficients and aggregate metadata. Source data terms remain separate from the MIT code licence.

Not clinically validated · Not for diagnosis, treatment, dosing or emergency decisions

Demonstration and submission readiness

09

RUN THE PROTOTYPE

- 1 Start Docker Compose**
- 2 Open localhost:3000**
- 3 Refresh the prediction**
- 4 Watch simulated CGM update the twin**

REMAINING TEAM ACTIONS

- Add the official team and college details
- Record and link the 20-minute demo video
- Complete the clinical language review
- Confirm the public repository from a clean computer

Code, tests, licence, data guide, model card and architecture source are ready.